

Assignment No. 07:

(Graph Theory)

BSCS 5B

Fall 2024

Due Date: January 01, 2025

Instructions:

- Each assignment should be typed and submitted in a clear, organized manner.
- Ensure all explanations are thorough and provide examples where necessary.
- Late submissions will incur a penalty unless prior arrangements are made.

Part 1: Conceptual Understanding

- Q1.** Define a graph in the context of graph theory. What are its primary components?
- Q2.** Explain the difference between a directed graph and an undirected graph, with examples.
- Q3.** What are adjacency matrices and adjacency lists? Provide a small example for both.
- Q4.** Describe the properties of a complete graph. How many edges does a complete graph with n vertices have?
- Q5.** What is a minimum spanning tree (MST)? List two practical applications.
- Q6.** Differentiate between breadth-first search (BFS) and depth-first search (DFS) in terms of traversal and usage.
- Q7.** What is a path in a graph? How does it differ from a cycle?
- Q8.** Explain the term isomorphic graphs. What does it mean for two graphs to be isomorphic?
- Q9.** Define the degree of a vertex in a graph. How is it computed for directed and undirected graphs?
- Q10.** State and explain the properties of a bipartite graph.
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Part 2: Applications and Analysis

- Q11.** Given the following graph adjacency list, write the corresponding adjacency matrix:
- A: {B, C},
B: {A, D},
C: {A, D},
D: {B, C}
- Q12.** Solve the minimum spanning tree (MST) for the graph below using Kruskal's algorithm. Include the steps involved.
Graph example: $G(V, E)$
where $V = \{A, B, C, D\}$, and

$$E = \{(A, B, 1), (B, C, 4), (A, C, 3), (C, D, 2)\}.$$

Q13. Describe how Prim's algorithm differs from Kruskal's algorithm in constructing an MST.

Q14. Explain the Bellman-Ford algorithm. How does it handle graphs with negative edge weights?

Q15. Write pseudocode for finding the shortest path in a graph using Dijkstra's algorithm.

Q16. Identify whether the following graphs are isomorphic. Explain your reasoning.

Graph 1: $V = \{A, B, C, D\}$, $E = \{(A, B), (B, C), (C, D), (D, A)\}$.

Graph 2: $V = \{W, X, Y, Z\}$, $E = \{(W, X), (X, Y), (Y, Z), (Z, W)\}$.

Q17. Solve the single source shortest path (SSSP) for a graph using Dijkstra's algorithm. Given graph:

$$V = \{A, B, C, D, E\},$$

$$E = \{(A, B, 4), (A, C, 2), (B, C, 5), (B, D, 10), (C, E, 3), (E, D, 4)\}.$$

Start from A.

Q18. Describe the difference between a spanning tree and a minimum spanning tree. Provide examples.

Q19. Prove or disprove: A connected graph with all vertex degrees even must have an Euler circuit.

Q20. Explain why the Graph Isomorphism Problem is in NP but not NP-Complete.

Part 3: Problem Solving and Critical Thinking

Q21. Design an algorithm to detect whether a graph contains a Hamiltonian Path. Discuss its time complexity.

Q22. For the following weighted graph, find the MST using Prim's algorithm starting from vertex A:

$$V = \{A, B, C, D, E\},$$

$$E = \{(A, B, 3), (A, C, 1), (B, C, 4), (B, D, 2), (C, D, 5), (D, E, 7)\}.$$

Q23. Solve the traveling salesman problem (TSP) using a brute-force approach for the graph $V = \{A, B, C, D\}$, with edge weights:

$$E = \{(A, B, 10), (A, C, 15), (A, D, 20), (B, C, 35), (B, D, 25), (C, D, 30)\}.$$

Q24. Given a graph with weighted edges, explain how to detect negative weight cycles using the Bellman-Ford algorithm. Illustrate with an example.

Q25. Discuss the implications of $P=NP$ on solving problems like Graph Isomorphism, Hamiltonian Path, and TSP. How would it change the computational landscape?